A Q-Learning-Based Routing Approach for Energy Efficient Information Transmission in Wireless Sensor Network

Xing Su, Yiting Ren, Zhi Cai, Yi Liang, and Limin Guo

***Abstract*—Nowadays, wireless sensor networks have played an important role in many applications. In these applications, a large number of wireless sensors are deployed in an environment to collect information and form network to transmit collected information to the base station or sink. Because wireless sensors have to work for a long period of time without maintenance, the energy consumption of wireless sensors has a great impact on the lifetime of wireless sensor network. To reduce and balance the energy consumption of wireless sensors, many information transmission routing approaches have been proposed. However, most of them do not consider all energy consumption factors of wireless sensors. To this end, an innovative information transmission routing approach based on Q-learning is proposed in this paper, which enables wireless sensors to adaptively select suitable neighboring sensors to achieve energy efficient information transmission in a decentralized manner. Based on the proposed approach, a wireless sensor first collects state and action information of its neighboring sensors, which includes information transmission direction and distance, the remaining energy, the energy consumption for information transmission and information transmission action. Then, all this information is used to update the Q-values of neighboring sensors, so as to enable the wireless sensor to select suitable neighboring sensor to transmit information according to Q-values. From simulation experiments, it can be seen that the proposed approach enables wireless sensors to reduce and balance the energy consumption of wireless sensors and extend the lifetime of the entire wireless sensor network.**

***Index Terms*—Wireless sensor network, Q-learning, local information, energy efficiency.**

I. Introduction

NOWADAYS, wireless sensor networks (WSNs) are applied to many fields, such as intelligent transportation [1], environment monitoring [2], military [3], etc. In these WSNs, hundreds of wireless sensors are deployed in target environments to perceive and collect information. After that the collected information was transmit to the based station or sink through WSN. By using WSN, people can obtain information of an environment without entering it. However, it also means that wireless sensors have to use battery to work without maintenance for a long period of time. Therefore, the energy of wireless sensors has a great influence on the working efficiency and effectiveness of WSN [4], [5], [6], [7], [8], [9]. How to reduce and balance the energy consumption of wireless sensors in a WSN during the information transmission are the main issues that need to be solved.

In the last twenty years, with the widespread applications of WSN, many approaches were proposed to reduce the energy consumption of wireless sensors from different perspectives [10], [11], [12], [13], [14]. To reduce the energy consumption of wireless sensors, some approaches enable wireless sensors to adjust their sensing or communication range [15]. Some approaches reduce the energy consumption of wireless sensors through allowing them to sleep and wake up alternatively [16], [17], [18]. In the early WSNs, wireless sensors directly transmit their collected information to the base station [19]. Since the energy consumption of information transmission is proportional to the square of information transmission distance [20], many routing approaches were proposed to reduce information transmission distance of wireless sensors. Routing approaches enable wireless sensors to transmit their information to the base station in a multi-hop manner with the help of their neighboring sensors [21], [22], [23]. In 2020, Titaev proposed an energy-saving routing metric for low rate sensor network [24], which considers both link quality and node initial energy values to evaluate the energy consumption of links among wireless sensors. In [25], Muruganathan et al. proposed a routing protocol through dynamic clustering, which can balance the energy consumption of sensor nodes and extend the lifetime of the WSN. Jin et al. proposed a centralized multi-hop routing approach [26], which uses multi-start minimum spanning forest to overcome the unbalanced energy dissipation and data conflict problems of low energy adaptive clustering hierarchy protocol. However, these routing approaches are mainly based on the fixed rules to find optimal information transmission paths, which are hard to handle dynamic changes of wireless sensors in WSN.

As a branch of machine learning, reinforcement learning (RL) were widely used in many research and application fields, such as game theory [27], recommendation system [28], automatic control [29], etc. In the last twenty years, RL began to

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be used to for wireless sensors routing in WSN. Centralized RL based routing approaches select information transmission paths based on the overall network topology. In 2021, Zhang et al. proposed a deep RL based routing approach [30], which uses double deep Q-network to make routing decisions for all wireless sensors in WSN. Abbood et al. also proposed a deep RL and SDN based routing approach [31], which uses deep RL based approach to balance the energy consumption of wireless sensors through discovering alternative paths to transmit information in the WSN. Decentralized RL based routing approaches enable each wireless sensor to select its neighboring sensors to transmit information with local information. Boyan and Littman [32] proposed a packet routing approach, which involves information delivery time and transmission distance into the Q-value calculation. Oddi et al. also proposed a Q-learning based routing approach [33], which considers residual batteries of wireless sensors and the control overhead. Li et al. proposed a Q-learning based routing protocol for underwater optical WSNs [34]. In their protocol, the residual energy and link quality are main factors of the Q-value. However, current RL based routing approaches also have some limitations, which are demonstrated as follows.

1) In many real WSN applications, it is hard to have a central controller in the WSN to collect state and action information of wireless sensors and find information transmission paths for them. In addition, the energy consumption for the global communication between the central controller and wireless sensors is huge;
2) Although based on centralized RL based routing approaches, the central controller can find optimal information transmission paths for all wireless sensors in the WSN, once there are any dynamic changes in the WSN, the optimal paths have to be adjusted or recalculated, the computational complexity for the process is high;
3) Although decentralized RL based routing approaches are adaptive to dynamic changes in WSNs, without the central controller to coordinate, many energy consumption factors need to be considered by wireless sensors during information transmission routing. However, most of current decentralized RL based routing approaches do not consider all these factors.

The motivation of this study is to overcome the limitations of current RL based routing approaches and use the advantages of decentralized RL algorithm to enable wireless sensors to achieve energy efficient information transmission in a WSN. In this paper, a Q-learning based routing approach is proposed, which enables wireless sensors to adaptively select suitable neighboring sensors to transmit information in a decentralized manner, so as to reduce and balance energy consumption of wireless sensors in the WSN. Specifically, for a wireless sensor, all energy consumption related state and action information of its neighboring sensors is incorporated into the Q-value update. The information includes the information transmission direction and distance, the remaining energy, the energy consumption for the information transmission and the information transmission action. Based on the Q-values of neighboring sensors, the wireless sensor can select the most suitable neighboring sensors to transmit information. The contributions of the proposed approach can be summarized as follows.

1) The proposed approach is a decentralized RL based routing approach, which does not need the central controller to coordinate for wireless sensors and saves the energy for the global communication;
2) The proposed approach is highly adaptive, which enables wireless sensors to adjust themselves to handle dynamic changes in the WSN;
3) The proposed approach incorporates all energy consumption factors into the reward function of Q-learning, which enables wireless sensors with local view to select most suitable neighboring sensors to transmit information;
4) The simulation experiments indicate that based on the proposed approach, all wireless sensors in the WSN can adaptively reduce and balance their energy consumption for the information transmission in a decentralized manner.

The rest of this paper is organized as follows. Section II gives the problem description and objectives of the information transmission routing problem. Section III introduces the basic principle of the proposed approach in detail. Section IV presents and analyzes the simulation experiment results. Section V introduces and compares the works related to the proposed approach. Section VI concludes the paper and outlines the future work.

II. Problem Description and Definitions

In an environment, m number of wireless sensors are deployed and form a WSN, which can be described as $SN = \{s_1, s_2, \ldots, s_m\}$, where s_i represents the i^{th} wireless sensor in the network. The definition of a wireless sensor can be described as follows.

Definition 1: A *wireless sensor* (s_i) can be defined as follows.

$$s_i = <sid, loc_i, reng_i, NSen_i>, \tag{1}$$

where *sid* is the identification of s_i in the WSN, $loc_i = (x_i, y_i)$ is the deployment location of s_i, $reng_i$ is the remaining energy of s_i, and $NSen_i$ is the set of all neighboring sensors of s_i, to which s_i can transmit information.

These wireless sensors collected information in the environment. After a period of time, wireless sensors transmit their collected information to the base station or sink (i.e., *base*). The location of *base* is $loc_{base} = (x_{base}, y_{base})$, which is known by all wireless sensors in the WSN. To reduce and balance the energy consumption for the information transmission, a wireless sensor (i.e., s_i) uses its neighboring sensors (i.e., $s_j \in SN$ and $s_j \in NSen_i$) to transmit information to the base station in a multi-hop manner. An information transmission of s_i to s_j at time period *t* can be defined as follows.

Definition 2: An *information transmission* of s_i at time period *t* $(it_{(i,t)})$ can be defined as follows.

$$it_{(i,t)} = <s_i, s_j, a_{(i,t)}>, \tag{2}$$

where s_i is the sensor transmitting information, s_j is the sensor receiving information and $a_{(i,t)}$ is the amount of information transmission between s_i and s_j at time period *t*.

According to [35], the energy consumption of a wireless sensor for the information transmission is proportional to the square of information transmission distance. In addition, the energy consumption of a wireless sensor for the information transmission is also proportional to the amount of information transmission. By using Euclidean distance, the energy consumption $(ceng_{(i,t)})$ of s_i for the information transmission (i.e., $it_{(i,t)}$) can be calculated based on [35], which can be described as follows.

$$ceng_{(i,t)} = E_{elec} \cdot a_{(i,t)} + E_{amp} \cdot a_{(i,t)} \cdot \Big((x_i - x_j)^2 + (y_i - y_j)^2 \Big), \quad (3)$$

where $E_{elec} = 50\ nano - joules/bit$ and $E_{amp} = 100\ pico - joules/bit/m^2$ are two constant coefficients, $loc_i = (x_i, y_i)$ is the deployment location of s_i, $loc_j = (x_j, y_j)$ is the deployment location s_j (see, Definition 1) and $a_{(i,t)}$ is the amount of information transmission (see, Definition 2).

By ensuring that all collected information can be transmitted to the base station, the objective of information transmission routing approaches is to reduce and balance the energy consumed by wireless sensors for the information transmission. Based on Equation (3), the objective of information transmission routing approaches can be described as follows.

The objective to reduce the energy consumption:

$$obj_{reduce} = min\left(\sum_{i=1}^{m} \sum_{\forall t} ceng_{(i,t)} \right), \quad (4)$$

where *m* is the number of wireless sensors in the WSN.

The objective to balance the energy consumption:

$$obj_{balance} = min\left(\sqrt{\frac{\sum_{i=1}^{m} (reng_i - \overline{reng})^2}{m}} \right), \quad (5)$$

where *m* is the number of wireless sensors in the WSN, $reng_i$ is the remaining energy of s_i (see Definition 1) and $\overline{reng}$ is the average of remaining energy of wireless sensors in the WSN, which can be calculated as $\frac{\sum_{i=1}^{m} reng_i}{m}$.

III. The Basic Principle of the Proposed Approach

The proposed approach enables each wireless sensor in the WSN to select suitable neighboring sensor to transmit information according to their Q-values, so as to reduce and balance the energy consumption of wireless sensors in the entire WSN. In this section, the basic principle of the proposed approach is introduced in detail. First, the overview of the proposed approach is described in Section III-A; Then, the process of neighboring sensor information collection is introduced in Section III-B; Finally, the process of Q-value update is introduced in Section III-C.

A. The Overview of the Proposed Approach

Generally, based on the proposed approach, the information transmission is an iterative process. In each iteration, there are two processes, which are 1) the neighboring sensor information collection and 2) Q-value update. In the neighboring sensor information collection, each wireless sensor (i.e., s_i) needs to collect the state and action information of its neighboring sensors, which includes the deployment location of neighboring sensors, the remaining energy of neighboring sensors after information transmission and the information transmission action of neighboring sensors. Based on the collected state and action information of neighboring sensors, s_i can update the Q-values of its neighboring sensors. Finally, based on the Q-values, s_i just needs to select the neighboring sensor with the highest Q-value to transmit information. If several neighboring sensors of s_i have the same highest Q-values, s_i just randomly selects a neighboring sensor from them.

B. The Neighboring Sensor Information Collection

By using decentralized Q-learning, after a period of time, a wireless sensor (i.e., s_i) in the WSN needs transmit its collected information to the base station. After information transmission, s_i needs to collect the state and action information of its neighboring sensors, so as to update the Q-values of neighboring sensors. The state and action information of a neighboring sensor (i.e., $s_j \in NSen_i$, see Definition 1) collected by s_i at time period *t* can be defined as follows.

Definition 3: The state and action information of s_j at time *t* can be defined as follows.

$$state_{(i,j,t)} =< sid, loc_j, reng_{(j,t)}, act_{(j,t)}, t >, \quad (6)$$

where *sid* is the identification of s_j in the WSN, $loc_j = (x_j, y_j)$ is the deployment location of s_j, $reng_{(j,t)}$ is the remaining energy of s_j after the information transmission, $act_{(j,t)}$ is the information transmission action of s_j, which can be 'back' or 'forward' transmission ($act_{(j,t)}$ will be explained in Section III) and *t* is the time stamp.

C. Q-Value Update

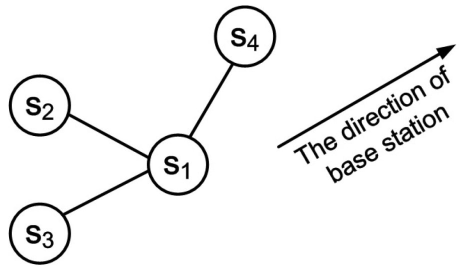
Based on the collected state and action information of neighboring sensors, a wireless sensor will update the Q-values of its neighboring sensors, so as to achieve energy efficient information transmission in a decentralized manner.

In the proposed approach, the Q-value (i.e., $Q_{(i,j)}$) of a neighboring sensor (i.e., s_j) of a wireless sensor (i.e., s_i) is related to five factors, which are

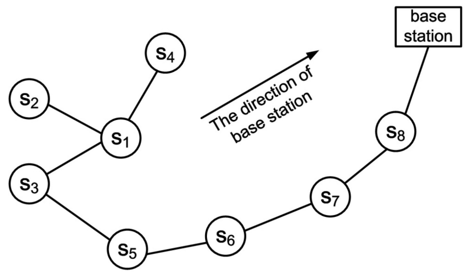
1) The direction difference between locations of s_j and the base station *base* (i.e., R_{dir_j});
2) The information transmission distance between locations of s_i and s_j (i.e., R_{dis_j});
3) The remaining energy of s_j after information transmission (i.e., R_{reng_j});
4) The energy consumption of s_j for the information transmission (i.e., R_{ceng_j});
5) The information transmission action of s_j (i.e., R_{act_j});

By considering all above factors, the reward function after last information transmission can be calculated as follows.

$$R_{(i,j)} = R_{dir_j} + R_{dis_j} + R_{reng_j} + R_{ceng_j} + R_{act_j}, \quad (7)$$



(a) The local view of a wireless sensor



(b) The global view of WSN

Fig. 1. The example of information transmission paths.

Based on the above five factors, the calculation of five factor rewards is introduced and explained in the following five sections.From Equation (7), we directly add five reward functions with the same weight, because this experiment is to calculate the five factors under the same status.If we assign them different weight values, it highlights which of the five factors is the most important and which is the least important.The reason why these five factors are set in this paper is that the wireless sensor adaptively selects neighbor nodes for forwarding under the influence of these five factors, so the weight values of these five factors are set to be the same.

1) The Direction Difference: The objective of this factor is to enable a wireless sensor to transmit its information to neighboring sensors closer to the base station, but also keep the opportunity to find the energy efficient information transmission paths through neighboring sensors far from the base station.

Since wireless sensors only have local view (i.e., they can only collect the state and action information of their neighboring sensors) about the WSN, during the information transmission routing, the wireless sensor close to the base station is not always the node on energy efficient information transmission paths. An example is shown in Fig. 1.

Fig. 1(a) shows the local view of a wireless sensor (i.e., s_1) during information transmission routing. From Fig. 1(a), it can be seen that s_1 has three neighboring sensors s_2, s_3 and s_4 and from the local view of s_1, the base station is close to s_4. However, from Fig. 1(b), it can be seen that the energy efficient information transmission path of s_1 is $\{s_1, s_3, s_5, s_6, s_7, s_8, base\}$. Although s_4 can directly transmit the collected information to the base station, according to the energy consumption equation (i.e., Equation (3)), the energy consumption of s_4 is huge, which increases the energy consumption of wireless sensors in the WSN.

The direction difference between locations of a neighboring sensor s_j and the base station *base* is calculated based on the angle (i.e., θ) between vectors (s_i to s_j and s_i to *base*), which is shown in Fig. 2

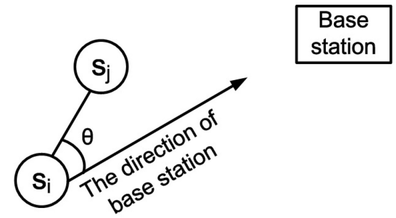


Fig. 2. The angle θ to calculate the direction difference.

Based on θ, the reward of direction difference of s_j (i.e., R_{dir_j}) can be calculated as follows.

$$R_{dir_j} = \begin{cases} 0^\circ \leq \theta < 45^\circ \ \ or \ 315^\circ \leq \theta < 360^\circ & 1 \\ 45^\circ \leq \theta < 135^\circ \ \ or \ 225^\circ \leq \theta < 315^\circ & 0 \\ 135^\circ \leq \theta < 225^\circ & -1 \end{cases} \tag{8}$$

For the calculation of R_{dir_j}, the concepts of quadrant is used, if the two vectors of θ are in the same quadrant, which means the direction difference is small, $R_{dir_j} = 1$, if the two vectors of θ are in two adjacent quadrants, which means there are certain direction difference, R_{dir_j} is 0, and if the two vectors of θ are in two opposite quadrants, which means the direction difference is big, R_{dir_j} is -1.

2) The Information Transmission Distance: Based on wireless communication technologies, the information transmission distance has a great influence on the energy consumption of wireless sensors in a WSN. In this paper, the information transmission distance between a wireless sensor s_i and its neighboring sensor s_j is the Euclidean distance between their deployment locations, which is calculated as follows.

$$dis(s_i, s_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}, \tag{9}$$

where $loc_i = (x_i, y_i)$ and $loc_j = (x_j, y_j)$ are the deployment locations of s_i and s_j, respectively (see Definition 1).

According to [20], the energy consumption for the information transmission is proportional to the square of the information transmission distance. The longer the information transmission distance is, the more energy the wireless sensor will consume. Therefore, the reward of information transmission distance of s_j (i.e., R_{dis_j}) should be inversely proportional to the square of information transmission distance between s_i and s_j, which can be calculated as follows.

$$R_{dis_j} = -\beta \times dis(s_i, s_j)^2, \tag{10}$$

where β is the coefficient to adjust the contribution of information transmission distance on the information transmission routing. The value of β is related to the settings of the other factors and experiments. In the proposed approach, the objective of β is to adjust R_{dis_j} to [0, 1], so as to be consistent with the rewards of other factors.

3) The Remaining Energy: One of the objectives of the proposed approach is to balance the energy consumption of wireless sensors in the WSN. The consideration of the remaining energy of neighboring sensors during information transmission routing is necessary. In the proposed approach,

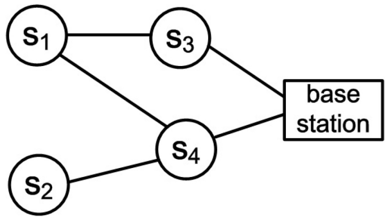


Fig. 3. The example of a WSN.

a wireless sensor s_i can directly collect the information of remaining energy of its neighboring sensors s_j (i.e., $reng_j$, see Definition 3).

Based on $reng_j$ of s_j, the reward of the remaining energy of s_j (i.e., R_{reng_j}) is calculated based on the percentage of remaining energy of s_j (i.e., $peng_j = reng_j / Eng_j$, where Eng_j is the initial energy of s_j), which is described as follows.

$$R_{reng_j} = \begin{cases} peng_j \geq thre,\ R_{reng_j} = peng_j \\ peng_j < thre,\ R_{reng_j} = -e^{peng_j} - E \end{cases}, \quad (11)$$

where *thre* is a remaining energy threshold to indicate percentage of remaining energy of wireless sensors, which is the average remaining energy of neighboring sensors of s_i (i.e., $thre = \frac{\sum_{s_j \in NSen_i} reng_{(j,t)}}{num}$, where *num* is the number of neighboring sensors of s_i) and E is a constant to adjust the contribution of the remaining energy to R_{reng_j}.

From Equation (11), it can be seen that the update of R_{reng_j} include two situations. When the remaining energy of s_j is no less than the threshold, R_{reng_j} is updated as the remaining energy of s_j; When the remaining energy of s_j is less than the threshold, R_{reng_j} is updated based on the exponential function, which can greatly reduce the chance that neighboring sensors with lower remaining energy are selected to transmit information;

4) The Energy Consumption for the Information Transmission: Unlike other information transmission routing approaches, the proposed approach considers the energy consumed by a neighboring sensor for the last information transmission. This is because that the proposed approach is a decentralized routing approach, where there is no central controller for the coordination and each wireless sensor in the WSN can only collect the state and action information of its neighboring sensors. For an information transmission, a wireless sensor s_i selects its neighboring sensor s_j to transmit information. However, s_i does not know how many other wireless sensors in the WSN also use s_j as the suitable neighboring sensor to transmit information.

As the example shown in Fig. 3. In Fig. 3, the circles and rectangle represent the wireless sensors and the base station. The solid lines between them indicate neighboring relationships of wireless sensors. From Fig. 3, it can be seen that s_1 has two neighboring sensors s_3 and s_4. Without considering other factors, it is the same for s_1 to transmit information to s_3 or s_4. s_2 only has one neighboring sensor s_4, so it must transmit information to s_2. If both s_1 and s_2 use s_4 to transmit information, the energy consumption of s_4 will be big. It is better for s_1 to select s_3 to transmit information. However, since s_1 and s_2 are not the neighboring sensors and they do not know the existence of each other, how to enable s_1 to know the existence of s_2 and select s_3 to transmit information is the main reason to consider the factor of the energy consumption for the information transmission.

The calculation of this factor is based on that the energy consumption for the information transmission is proportional to the amount of information transmission [35]. After received the information transmitted by s_i, the neighboring sensors s_j needs to add its own collected information to the received information, and transmit them to a neighboring sensor of s_j. We assume that the amount of information collected by wireless sensors in a period of time and the information transmission distance of wireless sensors are similar, so if s_j only receives and transmits information for s_i, the energy consumption for the information transmission of s_j should be twice the energy consumed by s_i to transmit information (see Equation (3)).

Based on the above assumption and principle, if the energy difference of s_j before and after information transmission is small (i.e., less than four times of energy consumed to transmit collected information in a period of time), it means that only a few other wireless sensors use s_j to transmit information. If the energy difference of s_j before and after information transmission is big, it means many other wireless sensors use s_j to transmit information. The energy difference of s_j before and after information transmission can be calculated by its state and action information in two continuous time periods (i.e., $state_{(i,j,t-1)}$ and $state_{(i,j,t)}$, see Definition 3), which can be calculated as follows.

$$\Delta reng_j = reng_{(j,t-1)} - reng_{(j,t)}, \quad (12)$$

where $reng_{(j,t-1)}$ is the remaining energy of s_j at time period $t - 1$ and $reng_{(j,t)}$ is the remaining energy of s_j at time period t.

Based on the energy difference (i.e., $\Delta reng_j$), the reward of energy consumption for the information transmission of s_j (i.e., R_{ceng_j}) is calculated as follows.

$$R_{ceng_j} = \begin{cases} \Delta reng_j > \Delta eng \ \ -1 \\ \Delta reng_j \leq \Delta eng \ \ 1 \end{cases}, \quad (13)$$

where Δeng is the energy consumption threshold, which can be calculated based on the energy consumption for a information transmission (i.e., see Equation (3)).

5) The Information Transmission Action: In addition to the state information of neighboring sensors, the information transmission actions of neighboring sensors can also provide useful information for information transmission routing. One of useful information transmission actions is back transmission. The back information transmission means that after a wireless sensor s_i transmitting information to its neighboring sensor s_j, s_j transmits the information of s_i and its own collected information back to s_i. Based on the proposed approach, there are two main situations that leads to the back transmission of neighboring sensors.

1) *The is no information transmission path* The training process of RL is long, during which each wireless sensor in the WSN has to continuously update the Q-values

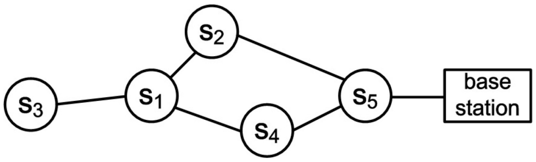


Fig. 4. The example of back transmission action.

of its neighboring sensors. At the beginning of the training process, the Q-values of all neighboring sensors are set to the same initial value, so each wireless sensor just randomly selects a neighboring sensor to transmit information. When a wireless sensor s_j selects a neighboring sensor s_j to transmit information and s_j happens to select s_i to transmit information, the back transmission action occurs.

2) *The information transmission path is not optimal* During information transmission routing, from the local view of a wireless sensor s_i, s_j is the optimal neighboring sensor to transmit information. However, from the local view of s_j, s_i is the optimal neighboring sensor to transmit the information to the base station. Through back transmission action, s_j is to tell s_i that currently, the information transmission path (through s_j) is not optimal.

As the example shown in Fig 4, a wireless sensor s_1 has three neighboring sensors s_2, s_3 and s_4. At the beginning of information transmission routing, s_1 randomly selects s_3 to transmit information. s_3 can only use s_1 to transmit information to the base station, s_3 adopts the back transmission action, which transmits the information of s_1 together with its own information back to s_1. After that, s_1 selects s_2 to transmit information, because the information transmission distance between s_1 and s_2 is the shortest. However, the information transmission path from s_2 to the base station is long. In some situations, such as the remaining energy of s_2 is low, s_2 can also adopt the back transmission action. Finally, s_1 selects the information transmission path through s_4 to the base station.

By considering the above two situations, when a neighboring sensor s_j adopts the back transmission action, the wireless sensor s_i has reduce the Q-value (i.e., $R_{(i,j)}$) of s_j to the second highest Q-value in all neighboring sensors of s_i. The reward of information transmission action of s_j (i.e., R_{act_j}) is described as follows.

$$R_{act_j} = \begin{cases} back & -\left|Q_{(i,j)} - Q_{(i,second)}\right| \\ forward & 0 \end{cases}, \tag{14}$$

where $Q_{(i,j)}$ is the current Q-value of s_j (i.e., the highest Q-value in all neighboring sensors of s_i) and $Q_{(i,second)}$ is the Q-value of the second highest Q-value in all neighboring sensors of s_i.

6) The Q-Value Update of a Wireless Sensor: Based on the proposed approach, the process of Q-value update of a wireless sensor (i.e., s_i) is described in Algorithm 1.

Algorithm 1: The Process of Q-Value Update of s_i

Input: neighboring sensors of s_i: $NSen_i$
Output: Q-values of $s_j \in NSen_i$
1 Initialize all $Q_{(i,j)}$ for $\forall s_j \in NSen_i$ with 0;
2 Set the learning rate α, β, E and Δeng;
3 **while** $reng_i \neq 0$ **do**
4   **for *each*** $s_j \in NSen_i$ **do**
5   Collect $state_{(i,j,t)}$ of s_j at time period *t*;
6   Calculate R_{dir_j} based on loc_j and loc_{base};
7   Calculate R_{dis_j} based on loc_j and loc_{base};
8   Calculate R_{reng_j} based on $reng_{(j,t)} \in state_{(i,j,t)}$;
9   Calculate R_{ceng_j} based on $reng_{(j,t)} \in state_{(i,j,t)}$ and $reng_{(j,t-1)} \in state_{(i,j,t-1)}$
10   Calculate R_{act_j} based on $act_{(j,t)} \in state_{(i,j,t)}$;
11         $R_{(i,j)} = R_{dir_j} + R_{dis_j} + R_{reng_j} + R_{ceng_j} + R_{act_j}$;
12         $Q_{(i,j)} = Q_{(i,j)} + \alpha \times R_{(i,j)}$;
13   **end**
14   Select s_j with $max(Q_{(i,j)})$ to transmit information;
15 **end**

Algorithm 1 can be explained as follows. The inputs of the algorithm is the neighboring sensors of s_i. The outputs of the algorithm is the Q-values of neighboring sensors of s_i. At the beginning, all Q-values of neighboring sensors are initialized to 0 (Line 1). The parameters are set, which include the learning rate, the information transmission distance coefficient (i.e., β in Equation (9)), remaining energy constant (i.e., E in Equation (13)) and the energy consumption threshold (i.e., Δp in Equation (3)) (Line 2). Then, the information transmission and Q-value update is an iteration process. The end condition of the iteration is that s_i does not have any energy (Line 3). In each iteration, for a neighboring sensor s_j, s_i first collects the state and action information of s_j (see Definition 3) (Lines 4 to 5). Then, based on the collected state and action information, the rewards of direct difference, information transmission distance, remaining energy, energy consumption for the information transmission and information transmission action of s_j are calculated (see Equations (8), (9), (13), (3) and (14)) (Lines 6 to 10). Finally, the Q-value of s_j is updated according to the rewards of five factors (Lines 11 to 12). When the Q-values of all neighboring sensors of s_i are updated, s_i will select s_j with the highest Q-value to transmit information (Line 14).

IV. Experiments and Analysis

In this section, the settings of experiments are first introduced in details. Then, four experiments are conducted and analyzed to evaluate the performance of the proposed approach, which include the average and deviation of remaining energy of wireless sensors in the WSN, the total energy consumption of wireless sensors in the WSN, the average of information transmission hops and the number of wireless sensors without energy in the WSN.

A. Experimental Settings

All experiments are conducted on a PC with 2.30GHz Intel Core i5 processor, and 4 GB memory. MATLAB R2020b is used to simulate the experimental environments. In experiments, 20 wireless sensors are randomly deployed in a

TABLE I
THE GENERAL SETTINGS OF THE EXPERIMENTS

Variables	Value
Number of wireless sensors	20
Size of environment	$100 \times 100\ m^2$
Maximum information transmission distance	50 m
Information transmission times	25000
Amount of transmitted information	3072 bits
Sensor initial energy	1 $joule$
Learning rate α	0.9

$100 \times 100m^2$ environment to form a WSN. The maximum transmission distance of wireless sensors is 50 *m*, so the coefficient in the factor value calculation of information transmission distance (i.e., β in Equation (12)) is set to 0.0002. The remaining energy constant (i.e., *E* in Equation (13)) is set to 1.222. The amount of collected environmental information that needs to be transmitted is 3 $\times$ 1024 *bits* = 3072 *bits*, so the energy consumption threshold (i.e., Δp in Equation (3)) is set to 0.001. The general settings of the experiments are shown in Table I

In the experiments, three information transmission routing approaches are selected as benchmarks to compared with the proposed approach, which are introduced as follows.

- SOWSN [20] is an energy-aware self-organizing routing approach, which uses some fixed rules to enable a wireless sensor to select neighboring sensors located in the area in the direction of the base station.
- UWSN [36] is a decentralized RL based routing approach, where each wireless sensor finds suitable neighboring sensor to transmit information by considering the remaining energy, the load and direction of deployment location of neighboring sensors.
- RLBR [35] is a centralized RL based routing approach, which finds optimal information transmission paths by considering the remaining energy, information transmission distance and the number of information transmission hops of wireless sensors.

B. The Average and Deviation of Remaining Energy of Wireless Sensors

In this section, during information collection and transmission, the average and deviation of remaining energy of wireless sensors based on the proposed approach and three benchmarks are compared and analyzed. In this experiment, the general experimental settings in Table I are used.

Since the objectives of information transmission routing approaches are to reduce and balance the energy consumption of wireless sensors in the WSN, the average and deviation of remaining energy of wireless sensors can directly reflect the performance of information transmission routing approaches on the above two objectives. The average and deviation of remaining energy of wireless sensors are calculated based on Equations 4 and 5 in Section II.

The experimental results of average remaining energy of wireless sensors are shown in Fig. 5, where X-axis is the information transmission times and Y-axis is the average remaining energy of wireless sensors in the WSN.

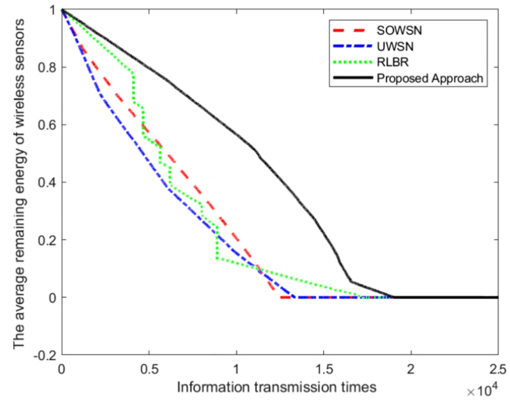


Fig. 5. The average remaining energy of wireless sensors.

From Fig. 5, it can be seen that in the first 15000 times of information transmission, the average remaining energy of wireless sensors based on SOWSN, UWSN and our approach decreases smoothly. However, in this process, especially in the first 8000 times of information transmission, the average remaining energy of wireless sensors based on RLBR fluctuates greatly. This is because that RLBR is a centralized RL based routing approach, which finds optimal information transmission paths for all wireless sensors in the WSN. Once there are any dynamic changes in the WSN, RLBR has to adjust or recalculate original information transmission paths, so RLBR needs a long time to find stable energy-efficient information transmission paths. From the overall change trends of average remaining energy of wireless sensors based on different routing approaches, it can be found that the average remaining energy of wireless sensors based on the proposed approach is always higher than that based on the three benchmarks, which indicates the advantage of the proposed approach on reducing the energy consumption of wireless sensors. This is because that during information transmission routing, SOWSN and UWSN do not consider the information transmission distance which is the largest energy consumption factor in the information transmission. After 15000 times of information transmission, since there are wireless sensors running out of energy in the WSN, the average remaining energy of wireless sensors decreases rapidly.

The experimental results of remaining energy deviation of wireless sensors are shown in Fig. 6.

The experimental results of remaining energy deviation of wireless sensors are shown in Fig. 6. X-axis represents the number of information transmissions and Y-axis represents the standard deviation of the remaining energy of wireless sensors.

From Fig. 6, it can be seen that the standard deviation of the remaining energy of wireless sensors based on the four approaches firstly increases and then decreases. This is because that based on the multi-hop information transmission, some wireless sensors in the WSN become forwarding sensors, which need to transmit information for other sensors. Therefore, the energy consumption of these forwarding sensors is much higher than that of non-forwarding sensors, which leads to an increasing deviation between the

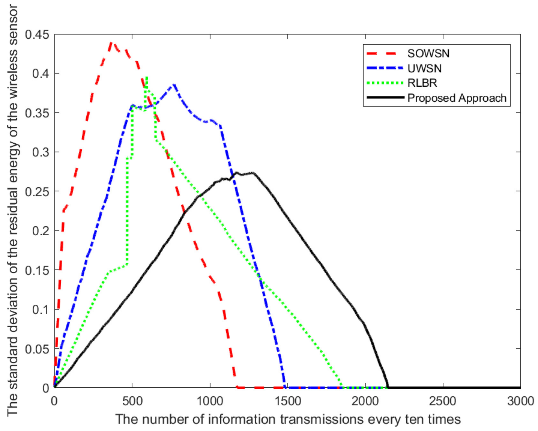


Fig. 6. The remaining energy deviation of wireless sensors.

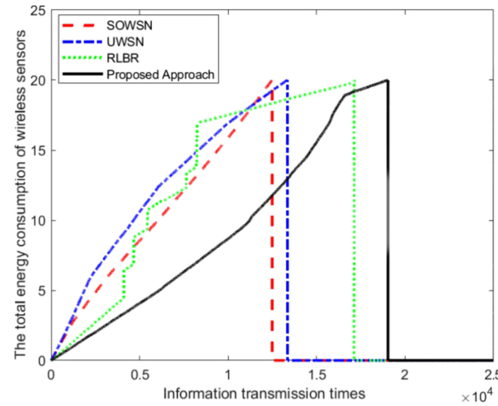


Fig. 7. The total energy consumption of wireless sensors.

remaining energy of wireless sensors in the WSN. After a period of information transmission, all wireless sensors find their optimal information transmission paths and wireless sensors alternatively become forwarding sensors to balance their remaining energy, so the deviation of the remaining energy of wireless sensors decreases. At the end, when all wireless sensors in the WSN run out of their energy, the deviation of the remaining energy of wireless sensors disappears.

Because SOWSN, UWSN and RLBR do not consider the balance of energy consumption of wireless sensors in the WSN, the remaining energy deviations of wireless sensors based on UWSN, SOWSN and RLBR are always higher than that of our approach. It indicates that based on UWSN, SOWSN and RLBR, although the energy consumption of wireless sensors can be reduced, they do not balance the energy consumption of wireless sensors in the WSN. The unbalanced energy consumption will cause some wireless sensors in the WSN, especially some sensors at important locations, run out of energy early and then leads to the rapid energy consumption of other wireless sensors in the WSN. Our approach considers the balance of energy consumption of wireless sensors during information transmission routing, so our approach can effectively extend the life time of WSN.

C. The Total Energy Consumption of Wireless Sensors

In this section, the total energy consumption of wireless sensors based on the proposed approach and three benchmarks are compared and analyzed. In this experiment, the general experimental settings in Table I are used.

Compared with the average remaining energy of wireless sensors, the total energy consumption of wireless sensors can more directly reflect the energy consumption of wireless sensors and the life time of the WSN.

The experimental results of total energy consumption of wireless sensors are shown in Fig. 5, where X-axis is the information transmission times and Y-axis is the total energy consumption of wireless sensors in the WSN. To clearly show the life time of the WSN, when all wireless sensors in the WSN are out of energy (i.e., their remaining energy is 0), the total energy consumption of wireless sensors is set to 0.

From Fig. 7, it can be seen that like the last experiment, for the same reason, the total energy consumption of wireless sensors based on SOWSN and UWSN and our approach increases smoothly, while the total energy consumption of wireless sensors based on RLBR increases fluctuatingly. From the overall change trends of total energy consumption of wireless sensors based on different routing approaches, we found that the total energy consumption of wireless sensors based on the proposed approach is always lower than that based on the three benchmarks. From the life time of the WSN based on different routing approaches, we can see that because SOWSN and UWSN do not consider the information transmission distance, the total energy consumption of wireless sensors based on them are high, so the life time of the WSN is short. After about 12000 times of information transmission, all wireless sensors based on UWSN run out of energy. The WSN based on SOWSN lives longer, which finish about 19000 times of information transmission. Although RLBR enables wireless sensors in the WSN to find optimal energy-efficient information transmission paths, the energy of wireless sensors consumes a lot during the long time learning and adjusting process. Therefore, after around 19000 times of information transmission, all wireless sensors in the WSN run out of energy. Unlike the benchmarks, based on the proposed approach, the WSN has the longest life time.

D. The Average Number of Information Transmission Hops

In this section, the average number of information transmission hops based on the proposed approach and three benchmarks are compared and analyzed. In this experiment, the general experimental settings in Table I are used.

The number of information transmission hops refers to how many times the information are transmitted from the wireless sensor that collects it to the base station. Since the energy consumption for the information transmission is proportional to the square of information transmission distance (i.e., see Equation (3) in Section II), the high number of hops indicates the short information transmission distance. However, with the increase of the number of information transmission hops, the amount of information that is transmitted will also increase exponentially. The suitable number of information transmission hops is the tradeoff between the information

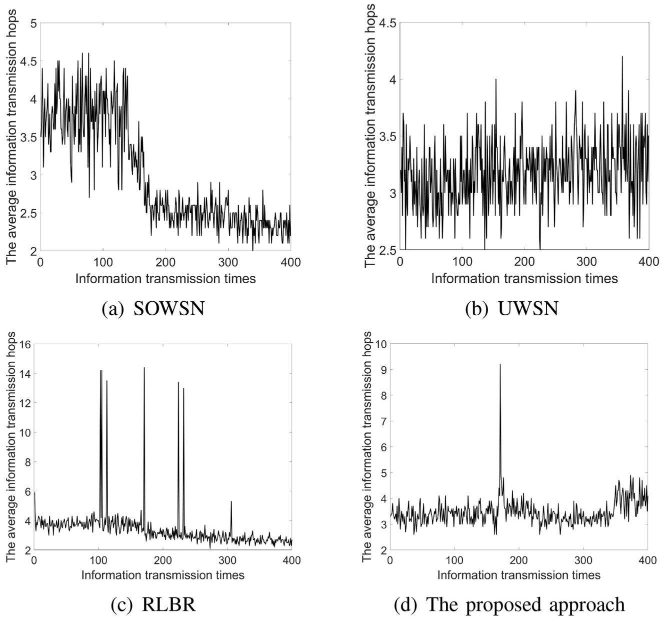


Fig. 8. The average information transmission hops of wireless sensors.

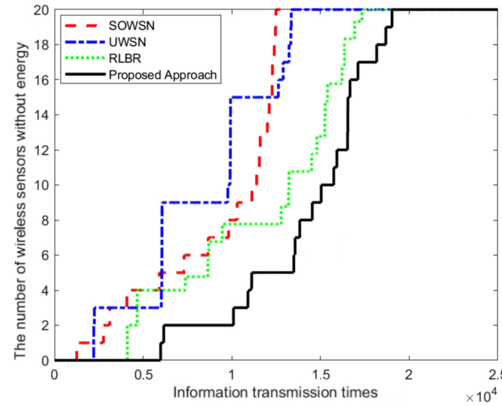


Fig. 9. The number of wireless sensors without energy.

transmission distance and the amount of information transmission.

The experimental results of average number of information transmission hops are shown in Fig. 8, where X-axis is the information transmission times and Y-axis is the average information transmission hops.

From Fig. 8(a) and 8(b), it can be seen that the average numbers of information transmission hops based on SOWSN and UWSN are relatively stable, which fluctuate between 2 and 4 hops. This is because that during information transmission routing, SOWSN and UWSN restrict the information transmission direction, that is only the wireless sensors in the direction of the base station can be selected to transmit information, so their numbers of information transmission hops are small. Although this limitation enables wireless sensors to find energy efficient information transmission paths easily, it also limits the adaptability of these two approaches. If there is no wireless sensor in the direction of the base station, wireless sensors will directly transmit their information to the base station rather than try to use wireless sensors in the other directions to transmit information. Therefore, it is difficult for SOWSN and UWSN to find the suitable number of information transmission hops in different structures of the WSN. Unlike SOWSN and UWSN, RLBR and the proposed approach do not have limitation on the information transmission directions and the numbers of information transmission hops are adaptively adjusted through the learning process. Therefore, from Fig. 8(d) and 8(c), we can see that there are several times that the average number of hops suddenly increases to around 10 hops. This is because that in the process of information transmission, due to the continuous changes of remaining energy of wireless sensors, the original information transmission paths are no longer energy-efficient paths, wireless sensors needs to adapt to find the current energy-efficient paths through RL. In addition, since RLBR is a centralized RL based routing approach, it is more sensitive to any changes of wireless sensors in the WSN, so it carries out more times of adaptations than the proposed approach.

E. The Number of Wireless Sensors Running Out of Energy

In this section, in the process of information transmission, the number of wireless sensors running out of energy based on the proposed approach and three benchmarks are calculated and compared. In this experiment, the general experimental settings in Table I are used.

The number of wireless sensors running out of energy can directly reflect the balance of energy consumption of wireless sensors in the process of information transmission and it can also indicate the adaptability of information transmission routing approaches to handle changes in WSN in some extends.

The experimental results of average number of hops are shown in Fig. 8, where X-axis is the information transmission times and Y-axis is the number of wireless sensors without energy in the WSN.

From Fig. 9, it can be seen that in the first 3000 times of information transmission, there are wireless sensors running out of energy in the WSN based on SOWSN, UWSN and RLBR approaches. However, based on the proposed approach, the first wireless sensor running out of energy appears at around 5000 times of information transmission. In addition, after the first wireless sensor running out of energy, wireless sensors continuously run out of energy in the WSNs based on SOWSN, UWSN and RLBR approaches. Based on the proposed approach, after twelve wireless sensors running out of energy, there are a certain period of time (i.e., about 5000 times of information transmission) in which no wireless sensor runs out of energy. So, we can say that based on the proposed approach, the adaptability of wireless sensors is better than that based on the three benchmarks. Based on the benchmarks, the time period for wireless sensors to run out of energy is in a big range, while based on the proposed approach, most of wireless sensors running out of energy in 11000 and 17000 times of information transmission, which indicate the good performance of the proposed approach on balancing the energy consumption of wireless sensors. After

20000 times of information transmission, the last wireless sensor in the WSN based on the proposed approach runs out of energy, while at the same time, all wireless sensors in the WSN based on the benchmarks already run out of energy. Therefore, the WSN based on the proposed approach has the longest life time in all WSNs based on different routing approaches in the experiment.

V. The Related Work

In the last two decades, many routing approaches were proposed for WSN to achieve energy-efficient information transmission from different perspectives [5], [6], [37], [38], [39], [40], [41], [42], [43], [44], [45]. According to the algorithms they are based on, the routing approaches can be roughly divided into rule and RL based approaches. In the following sections, the two kinds of routing approaches will be introduced and analyzed.

A. The Rule Based Routing Approaches

The rule based routing approaches achieve energy efficient routing through setting rules. Based on these rules, wireless sensors can find suitable neighboring sensors to transmit information.

Rogers et al. proposed a self-organizing routing approach [20]. In their approach, wireless sensors select the neighboring sensors to transmit information based on some fixed rules. That is, whether there is neighboring sensor in a certain area between the wireless sensor and the base station. If there is such a neighboring sensor, the information will be transmitted through the neighboring sensor. If there is no such neighboring sensor, the information will be directly transmitted to the base station. Based on these rules, the approach of Rogers et al. enables wireless sensors to automatically form the energy efficient information transmission paths.

Abdelhakim et al. proposed an energy efficient scheme for the real-time information transmission, called MC-WSN [46]. The characteristic of MC-WSN is that through active network deployment and topology design, the number of hops from any sensor to the mobile access point can be limited. This characteristic can ensure the energy efficient information transmission and make the network easier to control. Based on MC-WSN, the energy consumption for the information transmission of sensors can be greatly reduced.

Lipare et al. proposed a clustering and routing algorithm, which uses fuzzy logic to correlate the input values of the clustering with the routing algorithm [47]. During information transmission routing, the algorithm of Lipare et al. takes the remaining energy, the information transmission distance and the number of sensors in the communication range into account. Based on the above factors, the cluster heads and routers are selected and sensors are allocated to cluster heads according to the load capacity of cluster heads, so as to improve the efficiency of the information transmission.

Hajer et al. proposed a range-based opportunistic routing (ROR) protocol for WSNs [48]. ROR aims to minimize the forwarding sensors for each wireless sensor and the energy consumption of wireless sensors for the information transmission. During information transmission routing, ROR mainly considers four factors, which are distance, cost and error rate of information transmission as well as the remaining energy of wireless sensors. The experiments indicate that ROR outperforms other opportunistic routing protocols in terms of energy consumption of wireless sensors.

In general, rule based routing approaches are simple and easy to implement. According to pre-defined rules, wireless sensors can easily find the most suitable neighboring sensors to achieve energy efficient information transmission without long term learning process. So, the working efficiency and effectiveness of this kind of approaches are high. However, since the rules are pre-defined and fixed, the adaptability of rule based routing approaches is low. Once there are some situations in the WSN that rules cannot handle, it is hard for wireless sensors to achieve self-adaption. Unlike rule based routing approaches, the proposed approach employs RL to balance and reduce the energy consumption of wireless sensors. Although the learning process of RL needs a certain period of time to achieve near optimal information transmission routing, wireless sensors can adaptively adjust their information transmission path to handle dynamic changes in the WSN based on the reward function.

B. The RL Based Routing Approaches

The RL based routing approaches achieve energy efficient information transmission through a certain time of learning process. According to the learning object has global or local view about the WSN, the RL based routing approaches can be divided into centralized and decentralized RL based routing approaches.

1) The Centralized RL Based Routing Approaches: In the centralized RL based routing approaches, the learning object is the central controller, which has global view about the WSN, so it makes decisions for the information transmission routing for all wireless sensors in the WSN. Sharma et al. proposed a tailored Q-learning based approach for the energy efficient routing in the WSN [4]. The main objective of their approach is to use an improved Q-Learning algorithm to minimize the energy consumed by wireless sensors. Based on their approach, wireless sensors can efficiently route through data aggregation to learn the best routing strategy. To achieve multiple routing objectives, Wang et al. proposed a centralized RL based routing approach [49], called AdaR. AdaR can adaptively learn optimal routing strategy based on multiple routing objecives. In addition, AdaR employs least squares RL, which is data efficient and insensitive to initial settings, so it is suitable for the routing in ad-hoc sensor networks with the characteristics of unreliable communication and frequent topological changes. Younus et al. [50] proposed a software-defined wireless sensor network (SDWSN) controller, which optimizes the routing paths through reinforcement learning. At the same time, SDWSN includes four reward functions to optimize the performance of WSN by considering the global, average, global weighted and average global weighted rewards. Compared with other RL based approaches, SDWSN improves the life time of WSN by 23% to 30%. In [51], Akter and Yoon proposed an algorithm for duty cycle interval control based on RL in WSN. They apply the Q-learning to obtain the maximum duty cycle interval that supports various delay requirements and the given delay success ratio. In practice, only sink needs to calculate Q-leaning in the WSN. Experiments show that the algorithm of Akter et al. can effectively reduce energy consumption of sensors and extend the lifetime of the network while satisfying the required delay and delay success ratio.

The centralized RL based routing approaches enable the central controller to find optimal information transmission paths for all wireless sensors and handle dynamic changes in the WSN. However, to find optimal information transmission paths, the central controller needs a long term of learning and calculation. To have global view about the WSN, the overhead for the communication between the central controller and wireless sensors is also high. In addition, optimal information transmission paths are very sensitive to the dynamic changes in the WSN. Once there is any dynamic changes in the WSN, the central controller has to adjust and recalculate optimal information transmission paths. Unlike centralized RL based routing approaches, the proposed approach employs the decentralized RL algorithm, which enables wireless sensors to make decisions for the information transmission by themselves based on their local views about the WSN. Therefore, based on the proposed approach, the time for the learning process, computational complexity for the information transmission and the communication among wireless sensors can be greatly reduced. In addition, decentralized RL algorithm enables related wireless sensors to adaptively adjust to handle dynamic changes in the WSN.

2) The Decentralized RL Based Routing Approaches: In the decentralized RL based routing approaches, the learning objects are wireless sensors, which only have the state information about their neighboring sensors and they make decisions for the information transmission for themselves. Kim et al. proposed a RL based topology-aware routing approach for underwater WSN [36]. In order to enable wireless sensors to transmit information to the right direction, the approach of Kim et al. takes the local topology of wireless sensors into account. In addition, the approach of Kim et al. also considers the changes in channel status and the remaining energy of wireless sensors. Jafarzadeh and Moghaddam proposed a RL based energy-aware routing approach for WSN, called EQR-RL [52]. During the information transmission routing, the link quality and remaining energy of wireless sensors are considered. Their approach is extensible, since it does not need to maintain accurate network state information to make routing decisions. Instead, it handles the dynamic changes of the WSN by using RL. In 2020, Bouzid et al. proposed a decentralized RL based routing approach to achieve energy-efficient information transmission in the WSN [53]. In their approach, three factors related to the energy consumption of wireless sensors are considered, which are the information transmission distance, the remaining energy and the number of hops to the base station. Through dynamic path selection, the approach proposed by Bouzid et al. can achieve high energy efficiency and postpone sensors death and isolation. Al-Tous and Barhumi [54] proposed a distributed reinforcement learning based algorithm for the power control and information transmission in energy harvesting WSNs. The objective of the algorithm is to minimize the information transmission delay and the energy consumption of wireless sensors. In their algorithm, Al-Tous et al. employ a distributed state-action-reward-state-action mechanism, which can adaptively change the information transmission and power control actions of wireless sensors according to the states of WSNs. The experiments indicate that their algorithm has advantages in terms of power control of energy harvesting wireless sensors. In [55], Li et al. proposed a routing protocol based on multi-agent RL. The protocol takes the remaining energy and link quality into account during information transmission routing. In addition, two optimization strategies are proposed to accelerate the convergence of RL process. On this basis, a reward mechanism is provided for the distributed system. This protocol has certain restrictions on the structure of the network, where the average node degree is less than 14.

Without the coordination of the central controller, it is hard for wireless sensors to find optimal information transmission paths based on decentralized RL based approaches. In addition, to achieve near optimal energy efficient information transmission, many factors that affect the energy consumption of wireless sensors need to be considered. However, most of current decentralized RL based routing approaches do not consider all of them. Unlike these approaches, the proposed approach takes five factors related to energy consumption of wireless sensors into account. From experiments, it can be seen that the proposed approach can effectively reduce and balance the energy consumption of wireless sensors, so as to extend the life time of the WSN.

VI. Conclusion and Future Work

This paper proposed a decentralized RL based routing approach, which enable wireless sensors to select suitable neighboring sensors to transmit information with only local view about the WSN, so as to reduce and balance the energy consumption of wireless sensors for the information transmission and extend the life time of the WSN. In the proposed approach, five factors related to the energy consumption of wireless sensors are taken into account, which are the information transmission direction and distance, the remaining energy, the energy consumption for the information transmission and the information transmission action. These factors are involved in the reward function to update the Q-values of neighboring sensors. Based on the Q-values, a wireless sensor can select suitable neighboring sensor to achieve energy efficient information transmission. The simulation experiments indicate that the proposed approach can effectively extend the lifetime of WSN through reducing and balancing the energy consumption of wireless sensors in a decentralized manner. In the future, we intend to further balance the energy consumption of wireless sensors through taking more factors into account. In addition, we also try to use some prediction mechanisms to accelerate the convergence of RL.

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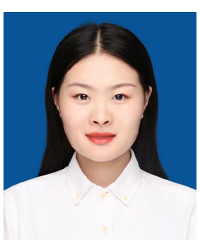

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